Production Deployment and Incident Response Procedures

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Department: Engineering

Team: DevOps

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1. Purpose

This document defines the deployment, monitoring, rollback and incident response procedures for production services at Meridian Technologies.

The DevOps team manages deployment infrastructure and coordinates production releases to maintain application availability and reliability.

2. Infrastructure Overview

Meridian uses the following infrastructure technologies:

Technology	Purpose
Docker	Application containerization
Kubernetes	Container orchestration
PostgreSQL	Relational database

Redis	Application caching
Kafka	Asynchronous messaging
Prometheus	Metrics collection
Grafana	Monitoring dashboards
Alertmanager	Alert management
Git	Source code management

3. Deployment Environments

The organization maintains three environments.

Environment	Purpose
Development	Feature development and testing

Staging	Integration and pre-production validation
Production	Live organizational services

Production credentials must not be reused in development environments.

Environment-specific configuration and secrets must be maintained separately.

4. Deployment Requirements

Before a production release, the responsible team must verify:

1. Code changes have been reviewed.
2. Automated tests have passed.
3. Dependencies have been checked.
4. Docker images have been built successfully.
5. Database migrations have been reviewed.
6. Staging validation is complete.
7. A rollback plan is available.
8. Production approval has been obtained.

Failed mandatory tests prevent production deployment.

5. Deployment Workflow

The standard deployment process follows six stages:

1. Source code integration.
2. Dependency installation and automated testing.
3. Docker image creation.
4. Staging deployment and validation.
5. Production approval and Kubernetes deployment.
6. Post-deployment health checks.

Production deployment requires approval from the responsible Engineering Lead.

Container images must use identifiable release tags rather than relying on mutable `latest` tags.

6. Deployment Health Targets

Metric	Target
Initial health check	Within 60 seconds
Deployment verification	Within 10 minutes
HTTP 5xx error rate	Below 1%
Normal CPU utilization	Below 80%
Normal memory utilization	Below 85%
Unhealthy replicas after successful rollout	Zero

These are internal operational targets and must be reviewed when service requirements change.

7. Rollback Procedure

A rollback is required when a release causes critical service degradation, data corruption or a significant security issue.

The responsible team must:

1. Stop further release promotion.
2. Identify the last stable deployment.
3. Notify the Engineering Lead.
4. Restore the previous stable application image.
5. Verify service health.
6. Check database compatibility.
7. Record the rollback event.

Database migrations require separate consideration because reverting an application image does not automatically reverse database changes.

Destructive schema changes must receive explicit review before deployment.

8. Incident Classification

Production incidents are categorized according to severity.

Severity	Description	Initial Response Target
SEV-1	Critical outage or major security/data impact	15 minutes
SEV-2	Major service degradation	30 minutes

SEV-3	Limited functionality failure	4 hours
SEV-4	Minor operational issue	1 working day

These targets represent initial response times, not guaranteed resolution times.

9. Incident Response

When an incident is detected, the on-call engineer must acknowledge it and create an incident record.

The record must include:

- Incident identifier.
- Detection timestamp.
- Affected service.
- Severity.
- Initial symptoms.
- Assigned responder.

The responsible team investigates application logs, infrastructure metrics and recent deployments.

Mitigation may involve restoring a previous release, restarting unhealthy workloads or recovering failed dependencies.

After service recovery, the incident record must include the resolution timestamp and actions taken.

10. Post-Incident Review

SEV-1 and SEV-2 incidents require a documented post-incident review within five working days after resolution.

The report must include the incident timeline, root cause, operational impact, mitigation actions and preventive recommendations.

Each corrective action must have an assigned owner and expected completion date.

11. Monitoring

Prometheus collects application and infrastructure metrics.

Grafana provides operational dashboards, while Alertmanager distributes alerts.

The DevOps team monitors service availability, CPU utilization, memory consumption, database connections, HTTP error rates and Kafka consumer lag.

Alerts must be reviewed periodically to avoid excessive false notifications.

12. Responsibilities

Role	Responsibility
DevOps Engineer	Deployment and infrastructure operations
Backend Engineer	Application debugging and code fixes
Engineering Lead	Production release approval
Incident Coordinator	Incident communication and escalation

13. Confidentiality and Access Control

This document is restricted to the DevOps team.

Authorized test identity:

- USR-00001 — Arjun Mehta.

Access to production infrastructure requires separate authorization.

Engineering department membership alone does not automatically grant production deployment privileges.

14. Related Documents

- ENG-001: Backend Platform Architecture and API Design
- ENG-002: Software Engineering Development and Deployment Guidelines
- ENG-003: Microservices Communication and Integration Standards
- IT-001: Enterprise Access Control Policy
- IT-004: Backup and Disaster Recovery Procedures
- IT-005: Security Incident Handling Playbook
- OPS-004: Incident Escalation and Internal Workflow

15. Version History

Version	Date	Description
1.0	2026-05-01	Initial deployment and incident response procedures

Document Owner: DevOps Lead